



AI Playbook and Toolkit for Public Health Departments

Cloud, Containers, and Secure Deployment Basics

ARC 330

Many AI tools are deployed in cloud environments or packaged in containers that make applications easier to move, scale, and maintain. Public health staff do not need to become cloud engineers, but they need enough technical literacy to understand hosting, identity and access management, logging, encryption, secrets management, backup, disaster recovery, and secure deployment responsibilities.

This module connects deployment choices to public health continuity, privacy, cybersecurity, and sustainability.

Level	intermediate/practitioner
Primary track	Technical Architecture Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of cloud, containers, and secure deployment basics in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a secure deployment readiness worksheet that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

Many AI tools are deployed in cloud environments or packaged in containers that make applications easier to move, scale, and maintain. Public health staff do not need to become cloud engineers, but they need enough technical literacy to understand hosting, identity and access management, logging, encryption, secrets management, backup, disaster recovery, and secure deployment responsibilities. This module connects deployment choices to public health continuity, privacy, cybersecurity, and sustainability.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Cloud, Containers, and Secure Deployment Basics matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

Cloud computing provides remote infrastructure, platforms, and services that can host data pipelines, analytics environments, AI tools, and public health applications. Cloud deployment can improve scalability and resilience, but only when identity, access, logging, encryption, and governance are configured correctly.

Containers package software and its dependencies so that it can run consistently across environments. Containers can improve reproducibility and deployment speed, but they must be scanned, patched, configured, and monitored. Container use does not remove the need for secure architecture.

Identity and access management defines who or what can access systems, data, and actions. In AI workflows, IAM must include human users, service accounts, APIs, agents, and automation tools. Poorly designed access can expose sensitive data or allow unauthorized write-back to systems of record.

Public Health Example

A public health agency may deploy an internal RAG assistant in a cloud environment so staff can search approved guidance and protocols. The tool may be low risk if it uses public documents, but it becomes higher risk if it retrieves sensitive policies, case notes, or internal operational documents. Secure deployment requires access controls, document-level permissions, logging, monitoring, backup, and a process for removing outdated content.

Technical Deep Dive

Secure deployment begins with environment separation. Development, testing, staging, and production environments should be clearly separated so experimental tools do not affect operational systems. Sensitive production data should not be copied into lower environments unless approved controls are in place.

Logging and monitoring are central to deployment safety. Agencies should know when systems are accessed, when APIs fail, when models produce errors, when unusual activity occurs, and when infrastructure changes are made. Logs should be protected because they can contain sensitive information.

Secrets management is essential for AI integrations. API keys, database passwords, certificates, and tokens should not be embedded in code, prompts, shared documents, or configuration files that are not secured. A compromised secret can allow unauthorized access to data or actions across connected systems.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a secure deployment readiness worksheet for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed secure deployment readiness worksheet. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

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Knowledge Check

- 1. What is the best reason to address cloud, containers, and secure deployment basics before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- NIST Cybersecurity Framework and cloud security guidance
- NIST Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- CDC Public Health Data Strategy and Data Modernization resources: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies